

IAB Tech Lab & Media Rating Council (MRC)

RTB Ad Fraud Compliance Book

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Updated & aligned with: app.py · rtb_data_generator.py · evaluate_system.py

IAB-AS-009 Ad Stacking	IAB-BF-100 Botnet Click Fraud	IAB-LS-200 Low-and-Slow Fraud	MRC-CTR-001 Format CTR Baselines
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SECTION 1

Objective & Operational Baselines

1.1 Purpose and scope

This document defines the technical and regulatory frameworks established by the Interactive Advertising Bureau (IAB) Tech Lab and the Media Rating Council (MRC) for identifying, categorising, and mitigating Invalid Traffic (IVT) within Real-Time Bidding (RTB) programmatic auctions. Compliance with these rules is mandatory for all supply-side networks, intermediaries, and publisher syndicates. This edition is updated to align with the RTB Fraud Detection System pipeline including three-tier fraud detection (IAB-AS-009, IAB-BF-100, IAB-LS-200) and format-aware CTR thresholds.

1.2 Baseline human engagement metrics

Metric	Compliant range	Flag threshold	Source in system
CTR — display_banner	0.05%–0.30%	above 0.80%	app.py CTR_THRESHOLDS
CTR — native_ad	0.20%–1.00%	above 2.50%	app.py CTR_THRESHOLDS
CTR — video_skippable	0.50%–2.50%	above 5.00%	app.py CTR_THRESHOLDS
CTR — rewarded_video	2.00%–8.00%	above 18.00%	app.py CTR_THRESHOLDS
Bid-to-Impression ratio	1.00x	above 1.30x	app.py Agent 1 filter
Win rate (legitimate)	70%–90%	below 15% suspicious	rtb_data_generator.py

Bid-to-Impression Equilibrium: One RTB auction win must equal exactly one rendered impression. This 1.00x ratio is the mathematical foundation of all impression-based fraud detection in this system.

2.1 Technical definition

Ad Stacking (also referred to as Impression Laundering or Pixel Stuffing) occurs when an ad-serving container script renders multiple ad creative components simultaneously within a single iframe layer, a hidden nested container, or a transparent 1x1 container. While only the top creative is visually perceptible to the human user, the underlying nested slots execute telemetry tracking pixels, extracting capital from media buyers for unseen impressions. Most commonly executed on **display_banner** placements (300x250, 728x90) because these formats use iframe containers that are straightforward to nest and hide. In this system, **pub_vortex_media** executes this fraud across rows 51–70 of the dataset.

2.2 Forensic detection thresholds

Signal	Threshold	In your data (pub_vortex_media)
Impression-to-Bid ratio (primary)	above 5.0x = CONFIRMED 1.3x–4.9x = investigate	ratio 5x–10x across 20 records (rows 51–70)
CTR on stacked inventory (secondary)	below 0.01% (invisible ads not clicked)	near-zero CTR — hidden ads cannot be clicked
Desktop browser prevalence (secondary)	90%+ desktop traffic	device_type = desktop, os_type = Windows/macOS
Agent 1 flag threshold	imp_bid_ratio above 2.0x	flag_reason = 'IAB-AS-009' on all 20 records

2.3 Financial damage formula

Variable	Formula	Example (pub_vortex_media)
Legitimate impressions	bids_won × 1	~9,200 per record
Fraudulent impressions	total impressions – bids_won	~55,200 extra per record (6x ratio)
CPM rate	\$2.50 (display_banner standard)	cpm_used = \$2.50 in dataset
Fraudulent spend / record	(Fraudulent impressions × \$2.50) / 1000	~\$138 per record
Total damage (20 records)	Sum of fraudulent spend_usd	~\$2,760 across all stacking rows

2.4 Enforcement actions

Upon IAB-AS-009 confirmation: (1) SSP executes immediate programmatic block on the compromised Publisher ID **pub_vortex_media** and all nested Ad Set IDs including **adset_hidden_premium_09**. (2) Media buyers entitled to instant financial clawbacks — all charges under fraudulent impression parameters nullified. (3) Publisher account suspended pending forensic audit.

3.1 Technical definition

Coordinated Click Fraud involves the deployment of non-human automated infrastructure — including headless web browsers, malware-infected device clusters (residential botnets), datacenter server loops, or automated macro scripts — to simulate genuine consumer interest by generating high volumes of synthetic click-through commands on ad payloads. Criminal operators specifically target **rewarded_video** (CPM \$8.00–\$15.00) because each fraudulent click earns 3.2x more than a display_banner click (CPM \$2.50). In this system, **pub_click_nexus** operates this fraud across rows 141–160.

3.2 Forensic detection thresholds

Threshold 1 — Format-aware CTR ceilings (matches app.py CTR_THRESHOLDS exactly)

Format	Human max CTR	Agent 1 flags above	Confirmed fraud above
display_banner	0.30%	0.80%	2.00%
rich_media_banner	0.50%	1.20%	3.00%
native_ad	1.00%	2.50%	5.00%
video_skippable	2.50%	5.00%	10.00%
video_nonskip	4.00%	8.00%	15.00%
rewarded_video	8.00%	18.00%	25.00% ← pub_click_nexus at 35–45%
interstitial	5.00%	10.00%	20.00%
push_notification	10.00%	20.00%	35.00%

Threshold 2 — Single-IP infrastructure exploitation

Traffic segments where more than 90% of cumulative click triggers originate from a single static IP address or private routing gateway constitute definitive Botnet Click Farm proof. Private IP ranges such as **192.168.x.x** indicate datacenter gateway routing. In your dataset: **pub_click_nexus ip_address = 192.168.1.100** (100% of all 20 records).

Threshold 3 — Device and OS signals

Botnet infrastructure operates on datacenter servers running **Linux**. rewarded_video is a mobile-primary format — legitimate audiences are 80%+ Android/iOS. pub_click_nexus shows **100% desktop Linux** on rewarded_video = triple-confirmed IAB-BF-100 SIVT evidence.

Threshold 4 — Auction velocity anomaly

Coordinated click scripts operate on lower-tier ad spaces, leading to depressed auction clearance velocity. Global Bid Win Rate crashing below 15.00% despite massive bid volumes sent constitutes a fourth corroborating signal. In your dataset, pub_click_nexus win_rate is 10%–15% (bids_won 1,000–1,500 out of bids_sent 10,000).

3.3 Enforcement actions

Upon IAB-BF-100 validation: (1) Hard-mitigation protocol triggered within 60 seconds — automated firewall block on **192.168.1.100**. (2) All ad request handshakes from **pub_click_nexus** and **adset_global_reward_clicks** dropped at edge gateway. (3) Immediate blacklist across all SSP routing tables. (4) Full clawback on all non-human click charges at \$8.00 CPM.

Keywords: impression ratio native ad subdued stacking shadow network low slow pattern fraud

4.1 Technical definition

Low-and-Slow Impression Fraud is a sophisticated evasion technique where a publisher deliberately maintains an Impression-to-Bid-Won Ratio between **1.3x and 1.9x** — below the standard automated detection threshold of 5.0x — while still generating fraudulent impressions. The impression ratio signal sustained above 1.3x across 10 or more records is the primary detection fingerprint. The operator accepts lower fraudulent revenue per auction win in exchange for remaining undetected by standard fraud engines. Predominantly executed on **native_ad** placements because native inventory blends into editorial content, making impression verification more complex than for display_banner. In this system, **pub_shadow_network** operates this fraud across rows 71–90 and 161–180 (40 records total).

4.2 Forensic detection thresholds

Signal	Threshold	In your data (pub_shadow_network)
Impression ratio (primary)	above 1.3x sustained across 10+ records	1.5x–1.9x across 40 consecutive records
Pattern duration	10+ records minimum for flag	40 records — far exceeds minimum
Agent 1 flag threshold	imp_bid_ratio above 1.3x	flag_reason = 'IAB-LS-200 suspected' on all 40 records
Geo diversity masking	US / IN / BR rotation	geo_country rotates across 3 countries to mask origin
Device masking	Mixed mobile + desktop	device_type = random mobile/desktop — avoids clustering
OS masking	Mixed Android / iOS / Windows	os_type rotated — unlike Linux-only botnet
Creative format	native_ad (fluid)	harder to audit than display_banner — deliberate choice

4.3 Severity classification

Ratio range	Classification	Action
1.00x (exactly)	COMPLIANT	None — passes Agent 1 clean
1.01x–1.29x	Minor anomaly — monitor	Passes Agent 1 (below 1.3x flag)
1.30x–1.99x	GIVT — IAB-LS-200 suspected	Agent 1 flags, Agent 3 investigates
2.00x–4.99x	IAB-AS-009 suspected	Agent 1 flags, Agent 3 escalates
5.00x+	IAB-AS-009 CONFIRMED — SIVT	Full enforcement

4.4 Financial damage formula

CPM for native_ad: **\$3.50** per thousand impressions (matches cpm_used in dataset). Fraudulent impressions per record = impressions – bids_won. Damage per record = (Fraudulent Impressions × \$3.50) / 1000. Net total = sum across all 40 flagged pub_shadow_network records.

4.5 Enforcement actions

IAB-LS-200 at GIVT level: (1) Formal warning to publisher. (2) Mandatory third-party impression verification tags on all native placements. (3) 90-day traffic audit of prior records. (4) Financial clawback on fraudulent impression portion (impressions – bids_won) at \$3.50 CPM. Escalation to SIVT if ratio reaches 2.0x or sustained pattern exceeds 30 consecutive sessions.

A single flat CTR threshold across all formats causes false positives (flagging legitimate rewarded_video at 6%) and false negatives (missing display_banner fraud at 3%). This section defines the format-specific thresholds that match the **CTR_THRESHOLDS** dictionary in app.py exactly.

Format (creative_type)	Ad size	Human max CTR	app.py flags above	Confirmed fraud
display_banner	300x250 / 728x90	0.30%	0.80%	2.00%
rich_media_banner	expandable	0.50%	1.20%	3.00%
native_ad	fluid	1.00%	2.50%	5.00%
video_skippable	1280x720	2.50%	5.00%	10.00%
video_nonskip	various	4.00%	8.00%	15.00%
rewarded_video	1920x1080	8.00%	18.00%	25.00%
interstitial	full-screen	5.00%	10.00%	20.00%
push_notification	notification	10.00%	20.00%	35.00%

Critical note: Fraud severity must always be assessed relative to the format ceiling, not a universal flat number. A 40% CTR on rewarded_video = 5x breach of its 8% ceiling. The same 40% on display_banner = a 133x breach of its 0.30% ceiling — infinitely more severe.

6.1 Legitimate human traffic distribution (2025–2026)

Device	Legitimate share	Primary OS	Note
Mobile	65–75%	Android / iOS	rewarded_video is 80%+ mobile
Desktop	20–30%	Windows / macOS	legitimate use
Tablet	3–8%	Android / iPadOS	legitimate use

6.2 Bot infrastructure fingerprints — matched to dataset publishers

Publisher	Breach	device_type	os_type	Why suspicious
pub_vortex_media	IAB-AS-009	desktop	Windows/macOS	Iframe nesting easier on desktop browsers than mobile
pub_click_nexus	IAB-BF-100	desktop (100%)	Linux	Datacenter servers run Linux. rewarded_video is 80%+
pub_shadow_network	IAB-LS-200	mixed (masked)	Android/iOS/Windows	Deliberate device rotation to appear organic — lack of
pub_premium_news24	CLEAN	60% mobile	Android/iOS/Win/Mac	Normal organic distribution — passes all device check

Triple-confirmation standard for IAB-BF-100: (1) CTR exceeds format ceiling per MRC-CTR-001 (2) 90%+ traffic from single static IP (192.168.1.100) (3) 100% desktop Linux on a mobile-primary format (rewarded_video). All three together = irrefutable SIVT.

MRC-CPM-00

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CPM Rate Standards & Damage Formulas — Section 7

Format	CPM range	Standard CPM	Used in dataset (cpm_used)
display_banner (300x250)	\$1.50–\$3.00	\$2.50	pub_vortex_media \$2.50
rich_media_banner	\$2.00–\$5.00	\$3.50	—
native_ad (fluid)	\$2.50–\$4.50	\$3.00	pub_shadow_network \$3.50
video_skippable (1280x720)	\$4.00–\$8.00	\$5.00	pub_premium_news24 \$5.00
video_nonskip	\$6.00–\$12.00	\$8.00	—
rewarded_video (1920x1080)	\$8.00–\$15.00	\$8.00	pub_click_nexus \$8.00 ← criminal target
interstitial	\$5.00–\$10.00	\$6.00	—
push_notification	\$0.50–\$2.00	\$1.00	—

Why criminals target rewarded_video

At \$8.00 CPM, 1,000 fraudulent rewarded_video impressions = \$8.00 stolen. Same 1,000 on display_banner = \$2.50. Criminals earn **3.2x more per impression** by targeting rewarded_video — explaining why pub_click_nexus specifically chose that format.

Damage calculation by breach type

Breach	Formula	Applied in system
IAB-AS-009 (Ad Stacking)	Damage = (impressions – bids_won) × CPM / 1000 Only fraudulent excess impressions charged	pub_vortex_media · CPM \$2.50 · 20 records · ratio 5–10x
IAB-BF-100 (Botnet)	Damage = sum of all spend_usd 100% fraudulent — no legitimate impressions exist	pub_click_nexus · CPM \$8.00 · 20 records
IAB-LS-200 (Low-and-Slow)	Damage = (impressions – bids_won) × CPM / 1000 Partial — only excess impression portion	pub_shadow_network · CPM \$3.50 · 40 records

8.1 Publisher registry

pub_vortex_media	SIVT — Confirmed Ad Stacking Syndicate
Fraud: IAB-AS-009 Impression Laundering	
Creative: display_banner (300x250) · adset_hidden_premium_09 · CPM \$2.50	
Device: desktop · Windows/macOS	
Signature: imp_bid_ratio 5x–10x · near-zero CTR · 20 rows (51–70)	
Action: Immediate blacklist · full clawback · adset disabled	
pub_click_nexus	SIVT — Confirmed Botnet Click Farm
Fraud: IAB-BF-100 Coordinated Click Fraud	
Creative: rewarded_video (1920x1080) · adset_global_reward_clicks · CPM \$8.00	
Device: desktop · Linux · 100% IP 192.168.1.100	
Signature: CTR 35–45% · rewarded_video human max 8% · 20 rows (141–160)	
Action: Immediate blacklist · IP 192.168.1.100 firewalled · full clawback	
pub_shadow_network	GIVT escalating SIVT — Active IAB-LS-200 Investigation
Fraud: IAB-LS-200 Subdued Impression Laundering on native_ad	
Creative: native_ad (fluid) · adset_native_content_07 · CPM \$3.50	
Device: mixed mobile/desktop · Android/iOS/Windows · geo US/IN/BR	
Signature: imp_bid_ratio 1.5x–1.9x sustained · 40 rows (71–90 and 161–180)	
Action: Escalated investigation · third-party verification · 90-day audit · \$3.50 CPM clawback	
pub_premium_news24	COMPLIANT — Verified Legitimate Publisher
Fraud: None — clean organic traffic	
Creative: display_banner + video_skippable · adset_standard_display_01 · CPM \$2.50/\$5.00	
Device: 60% mobile · Android/iOS/Windows/macOS · geo US/CA/GB/AU/DE	
Signature: imp_bid_ratio 1.00x · CTR 0.5–2.5% · 100 rows (1–50, 91–140, 181–200)	
Action: None — cleared by Agent 1 as clean	

8.2 Breach code quick reference

Code	Violation type	Primary detection signal	Detection in app.py
IAB-AS-009	Ad Stacking	imp_bid_ratio above 5x	Agent 1 flags above 2.0x
IAB-BF-100	Botnet Click Fraud	CTR above format ceiling	CTR_THRESHOLDS dict
IAB-LS-200	Low-and-Slow Fraud	imp_bid_ratio 1.3x–1.9x	Agent 1 flags above 1.3x
MRC-CTR-001	Format CTR ceilings	per-format human max CTR	CTR_THRESHOLDS dictionary
MRC-DEVICE-001	Device signals	device_type + os_type pattern	Agent 3 reasoning layer
MRC-CPM-001	CPM standards	spend_usd pre-calculated	rtb_data_generator.py
MRC-BASE-001	Publisher registry	known SIVT entities	evaluate_system.py index